

Active Inference Institute ~ 2023 Quarterly Roundtable #2

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Hello everyone, welcome to the Active Inference Institute quarterly roundtable number two in 2023. It's June 30th, 2023, and this is going to be our quarterly updates and ways that you can participate at the Institute and in the ACTIM ecosystem. So welcome to the Active Inference Institute.

We're a participatory online institute that is communicating, learning, and practicing applied active inference. You can find us at the links on this page and note that we'll be transitioning our emails over from activeinference at gmail.com towards activeinference.institute domains. This is recorded in an archived live stream, so please provide us with feedback so that we can improve our work. All backgrounds and perspectives are welcome and we'll be following video etiquette for live streams. Today in the quarterly meeting number two of the year, we'll start at the Institute scale, give some updates at the Institute level, and then we'll look a little bit deeper into primarily education but also research. And then I'll have some closing thoughts by way of a presidential address about where we're at with active inference. And if you're watching live, please write comments and questions in the live chat and we will look to address those during. Okay, so starting with the Institute scale, some updates coming from the Institute.

Here's one way that we can visualize the Active Inference Institute in June 2023. We are within the active inference ecosystem as well as within other ecosystems. And we have a nested structure. Alex, would you like

to describe this? Yeah, I think for this moment, it's great to see this structure and understand what that is actually in place and working and acting and adapting. And from very beginning, we consider it from perspective of systems thinking and from perspective of active inference, we always think about some kind of mastect and hierarchical structure which can be builded and structured accordingly. So having it right now, like embodied, it's great to see it.

Totally agree. At the Institute scale, where these first updates are going to come from, we have the officers, the board of

directors, and the scientific advisory board. We have two organizational units, which steward the development and

support participation in projects. And projects and learning groups are really where it happens on the ground. And so we're

going to talk a lot more about these, but this is the nesting and our own relationship with the ecosystem.

Another way to see this is in terms of people and positions at the Institute. So starting from the left, we have casual

engagement with our products and services into the niche. So this might be the very video that you're listening to or

watching right now, it could be the transcript of this video that you're reading, it might be the active inference ontology, it might be the active blockference software package, there's all these different open

source

products and services that we provide to the niche. And one moment, you will decide to participate in a project or a learning group. And that can be more towards the learning more towards the doing and leveraging your

contributions at the Institute. And some participants will choose to solidify their participation through volunteering or engaging with the internship program, both of which are new and really excitingly developing well. And then we haven't listed any names of the many of you who are in all of these different stages. And you may want to play another role or another position in the Institute, such as the Scientific Advisory Board, Board of Directors or the officers. So a lot of appreciation and thanks to everybody who's shown and not shown on this slide. We are really appreciative of how you all show up for the

Institute. Alex, want to add anything on that?

Yeah, even from this slide, we can see that a lot of people already involved in activity and provide great insights and different perspective and backgrounds and experience. And yes, a lot of people not showing here, but we

know them through our community platforms and so on. So you can see this progress in terms of participation in activities and it's also great progress.

All right, moving on, we are excited that for our third applied active inference symposium, to be held in August. We will have the theme of enacting ecosystems of shared intelligence, active inference as a cognitive kernel. We have a really broad and global co-organizer team, which is very exciting. So thanks to everybody who's in the co-organizing team for this symposium, we're just getting started on the organization. So in the coming days and weeks, we'll be providing more information on participating and a more formal call for presenters. However, if you're listening to this message now, and you are really excited to do a presentation, or a workshop or mentorship, and be a part of the active inference ecosystem, then please reach out to us. Alex, anything you want to add here?

Alex, anything you want to add here?

But actually, yes, I think we can remember that it will be a third symposium. And it's cool that we have that yearly rhythm for this type of activity. And hopefully this event will bring us and for people who will participate in new insights, connections and ground for future developments in their research.

Cool. All right, just to summarize and update a few of the Institute level programs. So volunteering at the Institute,

it's a way to engage, it is just making it that much easier to get involved with one or more projects and to start to get that

recognition from the Institute, understanding your contributions and also on your own resume, your own work history or your

portfolio. Volunteering is really an incredible way to learn by doing. If you want to apply or learn about organizational

active inference, or if you want to think about communication as active inference, education in the context of active

inference, gripper and the gripped. If you want to perspective swap and see it multiple ways, learning by doing on a

project through volunteering, we hope is a really unique opportunity. And what you do will and does really matter. So if you're taking notes on the margin of some paper, or if you're developing lists that you felt like help your onboarding into active inference, you can make a leverage contribution by sharing that kind of information. Or if you're not quite sure what type of contribution or the impact it would have, we can help find that

kind of a niche for you. The internship program is being a volunteer, but wanting a little bit more structure around that. We have about 20 currently active interns from all around the world, different backgrounds, levels of technical familiarity with the active inference, ranging from truly some of the people who I believe are experts to people who just found out about active inference, and wanted to use this program to accelerate their learning a lot. Also, in the last weeks and months, I would like to commend the two

individuals who know who they are, who have completed their internship. This has really been a great honor to

work with you, both of you. And to those who are still active, I look forward to seeing those updates and continuing to work with you. The internship consists of learning, doing, and also mentorship and facilitation.

Interns tend to participate in one or more course or textbook group, or maybe help out with some live streams. Interns engage on projects, and that is usually a minimum of two projects. One is hosted at the institute, so some of the projects that we'll look at later in this roundtable, but then also most of the interns have a personal or background research agenda. So sometimes they make very small contributions

to an institute project, but they're working really diligently on their dissertation or on their hobby project. Other times, it's most of the contribution to the institute project and then a little bit of self-guided research. And we try to provide our interns with support over multiple spatial and temporal scales. In terms of our engagement in the broader digital cyber-physical niche, we continue to grow on our different platforms and channels, though these are just proxy measures in our discord, podcast, twitter, and youtube, as well as mailing lists. So we continue to communicate out to those who

want to hear.

Here's the big picture on institute activities, and you can find this information at the activities page that's listed here. At the top, we have type of project, publication, live streams, such as the one happening right now, and the active inference journal, which picks up when the live stream hits stop, and transcribes and curates, publishes that transcript, which we'll talk more about. So live streams and journal are a great pair. In the active blockference project, we continue to build the soft and hard skills associated with generative modeling in the active inference tradition. And so we use the complex adaptive systems modeling framework, CAD-CAD, to implement different active inference generative models. This is a really great place, a lot of discussions about modeling. Robotics and embodied, JF, Codger, continues to offer his expertise and updates from this really incredible project where real

LEGO robots are implementing symbolic active inference. So thanks, JF, for hosting this project the way you do.

In the active inference ontology, we continue to refine and update the active inference ontology, which is the real language that we speak, whether it happens to be deployed in English or visually or in a programming language or however, the active inference ontology is how we're communicating, and we develop that and offer it as a service to the niche in this project.

We have multiple ongoing learning groups, which we'll talk more about soon, but we have a textbook group with Parr et al., the 2022 active inference textbook. We have Sanjeev Namjoshi, who's writing a really incredible work that will be a textbook published as Fundamentals of Active Inference, and we're alpha testing along with Sanjeev and right there alongside him. So thanks, Sanjeev, for bringing this work in progress to the Institute and to really sticking with the trouble.

This is going to be an incredible product. And we have two courses, one on physics as information processing and one on active inference for the social sciences, which we'll address soon. And also multiple times per week in our Discord, we have open discussions for education and research.

So if this is all a lot or you're not sure about anything, probably the best way to just drop in and find out would be to check out when and how we're meeting and come to the open discussions if you just want to like see what's up. Anything you want to add there, Alex?

Good to see that projects are active and mostly maybe for last period of time, it's having that new courses, which we'll describe later, but having and feeling it's growing. It's provide great opportunities for second half of the year.

So true. In the first quarterly roundtable, we had no courses. But as of now, we can say that there are two courses that are really pushing the frontier of education and even research in quantum information sciences and in social sciences. So thanks again. We'll get to those soon. All right, let's dive into a little more detail on EduActive. So Alex, would you like to first introduce and give a little thoughts on the Active Inference Qualification System draft?

Yeah, we put it here like for now at education, EduActive scale, but in future it will be applied for all institute structure and activities. We should have some cornerstone in terms of how we can measure effectiveness of our educational activities and educational services. In the beginning to understand how we can measure effectiveness of our educational activities and how we can measure effectiveness of our

courses and our materials, how we can change actually someone's mind and how deep this mind needs to be changed

in some formal sense. And we take, we're not started from Greenfield, we just take one of a stable system of qualifications level from European and adapted for how we can see it in application to Active Inference and participations in projects and so on. And for now we have a basic draft and we'll have discussions around these interested parties on different scales of institute and finally have some some stable form which will apply to different directions.

Thanks. I'll just add a few notes on this first, just echoing your last point. If you're interested

in Active Inference professionalization, education, qualification systems, even as a critic, we would really love to have you in the game. So please participate early, participate often. And I'll just highlight a few pieces of this system draft that are really exciting. First, it begins at level zero, which everybody de facto can have, perhaps even upon or before birth. Then as the levels proceed, active inference is always being connected with domains and systems of interest. So it's about yes and with active inference about connecting active inference to other fields and topics, understanding a different domains, ontology, robotics, or neuroscience, or pure mathematics, whatever it may be, and working with active inference as you gain knowledge of the ontology and of the details of active inference. But it's not just a knowledge-based qualification system. We also are looking to integrate skills and even project responsibility and autonomy. So just to give kind of an example of this, levels one and two, again, currently very tentative drafts, so for your engagement, but in levels one and two, people are able to act as basic learners and collaborators and begin to get a grasp on what terms we're using in active inference and how do those concepts relate to their field of interest. And we see this every single day in the textbook group and in discussions, someone says, oh, I'm working as a physical therapist. And so in action, that's like the person moving their arm. That is where these levels are targeted. Level three begins to get into describing and understanding some of the technical details of what we do. Level four begins to approach the evaluation and recapitulation of technical models in active inference. Level five is related to comprehensive development and implementation, as well as in the projects and autonomy, exerting more supervision for self and others. Level six, developing advanced skills. Level seven, engaging with very specialized problem solving skills. And level eight, engaging in synthesis, evaluation, and critical problem solving. So again, can't emphasize enough, it's just a draft. And we are calling for your participation if this is something that you would be interested in using or engaging with. But we think this is going to be a really important direction for the active inference ecosystem.

Carrying on to the textbook group. So in the active inference textbook group, we are working on and through the PAR, Pizzullo and Friston 2022 textbook. Currently, we have cohort three working on the second half of the book and cohort four on the first half of the book. Together, these two cohorts compromise about 150 people. Not every person joins every meeting. Some people engage with the recordings asynchronously. Other people are in our shared epistemic niche, just gleaning and contributing to this kind of collaborative, stigmatic process. As educators, we're focused on the continual delivery and deployment of questions and enrichment and discourse. And just some ways that this plays out. Here are, for example, questions that people have asked live or before or after a synchronous meeting, where we can connect the specific figures and the ontology terms to people's uncertainties. And on the right is an example of an equation. Here's the variational free energy equation 2.5 and the way that we can use the ontology to unpack and connect some of the dots with these equations. And in the last few weeks, we've actually taken that even further and explored how to use language models through Coda AI in this process. So here we have equations 2.1 and equations 2.2. They are being rendered in Coda via typing in LaTeX.

And then we are using language models within Coda to provide explanations, contextualizations for different backgrounds and expertises, and also translation into different languages, describing the variables and the notation and the who, what and why. So this has been a pretty fun avenue and a really cool example of how language models and education can be hand in hand and empower our own learning by using all kinds of models to help us where, frankly, it can be quite challenging, which is in understanding some of the analytical details of the framework. Anything you want to add, Alex?

Alex. This is the type of technology which looks like magic and but and and but having in place ontology which is nicely integrated for Coda again to all projects. It's really like an engine for this car which is getting more and more speed.

Alex. Cool. Okay. In May 2023, we began our first course at the Institute and that was Physics as Information Processing being taught by one of the leading researchers in the area, Chris Fields, and with a math PhD course assistant, Ander Aguirre. So, Chris, thank you for stepping up again from the Scientific Advisory Board to offer this truly incredible and unique educational experience. We've had massive response, dozens of registrants who want to engage in the participatory discussions, that happen in between your lectures, the second discussion being tomorrow for those who want to join, and 1500 views of the first lecture in a month. There's many people who are following along.

The course syllabus is explicit and innovative. We have a question and answer system so that people can ask questions asynchronously. Chris then answers them and we have them posted again through this interactive Coda interface and that's really supporting an incredible level of discourse and participation. All transcripts are going to be curated and published along with the questions and answers and some other supplemental information. So, this has been a really great experience and something new over the last few months.

Chris Fields, anything you want to add? I'm tempted to just talk about how cool the quantum information free energy principle is, but perhaps now is not the moment.

Chris Fields, anything you want to add to this?

For me, it's like historically from my whole background, starting from beginning on this century, starting from IT technology, technologies and computer science and this notion of information processing. I always was like engineering in engineering, like working field and working domain.

But I always had that need and feeling for first principles and for physics in place. And for now, seeing that latest developments on quantum information theory and how it's nicely connects to free energy and active inference, it's really, really great to have these frontier curves in place in our activities. And really great thanks to Chris and Ander.

And I believe that it's a kind of starting point, maybe 10 years after we'll come back all together and discuss how this course started great changes for different ecosystems.

Thank you, Alex. Totally agree.

And also, we are fresh off the presses, happy to announce the second course that we'll be offering at the Institute, which is Constructing Cultural Landscapes, Active Inference for the Social Sciences. And this is going to be a multi-teacher, massively participatory course.

Avel of Kairos Research, thank you for taking the initiative to suggest even this possibility.

And Ben and Mal, Lorena, thanks a lot for joining on for the journey. On July 7th, we will begin.

And it will have some lecture sessions as well as participatory discussion sections. All sessions are free. They'll be live streamed and rewatchable. The transcripts will be curated and published into a book at the end of the year. And this is going to be a really great on-ramp and space for people with a humanities or social sciences background. So we've heard people waiting for such a thing. It was Avel's insight that such a course could be hosted at Kairos Research and at the Active Inference Institute. So thanks to all the teachers here and just really looking forward to how you tackle and engage with this question of active inference in the social sciences. Anything you want to add?

Yeah, like these two different courses, like from different to distinct scales, like from quantum scale and social scale. And possibly it's like two different ends of spectrum in which we can work on with our research and education. It's great that this step to social sciences, because it brings a active inference and its formal and first principles principles framework into the domain which haven't got it before. And now it's a great beginning also. And our scales between quantum and social will be on us to work on next years.

Yes. And the kind of Ouroboros or horseshoe theory with quantum social sciences.

Onto the live streams. So, a lot to say. Of course, a little bit meta since we're on a live stream right now. But I'll just note a few things. First, across our different series, we now have 336 completed or planned live streams. So, most of those have already happened, but there are several that are planned, but we're over 300 completed and 336 as of now being planned. And even just specifically looking at the next 30 days, we have just such a breadth and depth. It's quite exciting to watch, let alone to enact. We'll have the Chris Fields course, a discussion section, a lecture, and then another discussion section. We'll begin the active inference for the social sciences course with Avel, Ben having contributions. We'll have a guest stream with professors from around the world, researchers, all these different topics, and all these different topics being able to be slotted in with a really great way to platform and give visibility and catalyze discussion around these incredible researchers work. So, it's really exciting. And it would be great. Not going to say it's specifically a personal request, but if you choose to read it as such, then great. It would be awesome to have more assistance with participating and preparing and enacting live streams. We have role-based ways of planning and preparing and enacting live streams. So, whether you want to play a more technical or operational or logistical role, if you want to be a conversational facilitator, if you want to do it in a different language, if you want to do a series, there's so many ways that you can get involved, but it would just be awesome to have more voices on the live stream and just be able to continue offering these kinds of resources at even faster rate with increased professionalization. And I know that there's many people out there who would be incredible contributors to broadcast and or facilitate. Everyone is also welcome to self-nominate or suggest or introduce somebody who you think would fit well on a stream. If it's like a perspective that we haven't heard from, that would be really important. There's a lot of background technical and educational work on the .zero videos when we do a

paper discussion. So, if there's a paper that you want to have a million x leveraged contribution on your preparation around, if we discuss that in a live stream and you're able to help before with the .zero, you'll be a co-author on a research publication of the transcript and you'll have made an absolutely unique irreplaceable contribution to active inference knowledge with what you did in your diligence in the .zero.

So, for people who hear that calling, that's incredibly important.

We invite people to share the live streams on social media or with their colleagues, however they find it relevant. Like if there's a topic or a person that has come up on our streams, sharing it where relevant will help bring people in. And also, we have weekly live stream meetings. So, if you just want to find out like what we do and how we do it in the live streams, then I would recommend that you join the regular meetings. And if you really want to join, but that time never works for you, then email and let us know that you're specifically interested. We'll find a different time. Any other comments there, Alex?

Alex Kovacic

Alex Kovacic

For this moment, we can see that

a lot of live stream activity is happening with participation of people, not only with our own preparations, but with people who just want to share ideas and so on. And if we can keep this rhythm to have different people with their interests and sharing ideas. And yes, if people will join to help us to do it more and more often, maybe it will be great.

Alex Kovacic

Thank you.

The Active Inference Journal picks up where live stream stops. The Active Inference Journal is engaged in continuous deployment and development of speech to text transcription, automated and manual editing curation. And finally, publication. So just to give an example, a live stream will occur. And then we use speech to text onto a public GitHub repository, everybody can edit the markdown file. And then we're able to publish those transcripts so that they can be citable, so that a conversation someone has can go onto their resume as a true citable publication. And we can just facilitate the accessibility and the findability of these incredible conversations and moments that we have. And also in the Active Inference Journal, we're always developing the process from setup, for those of all technical backgrounds, how to get things working on your own local setup or on a cloud setup using free and open source software where possible. The speech to text, the text processing, the editing, translating into languages, rendering into a diversity of formats, publishing through different channels, and then engaging in further analysis related to natural language processing, and the Active Inference Ontology, all these fun things we can do. So if you, in an alternate multiverse, were a librarian, or an editor, or a reviewer, or an advocate of the Active Inference Corpus, bring it into this reality and make a huge contribution by cultivating our epistemic garden and our offerings because there's so much enrichment that can happen

with past streams and with proactively letting the live stream project know and letting courses know how we can help them at the

journal to develop these kinds of publications that in new ways are expanding definitions of scholarship and research and service.

anything else to add?

yeah, as for this project, I believe it's like most technical heavy in the institute and maybe we can call it more broadly like a language processing project.

and maybe it is a good starting point for people who have good technical skills, some programming skills which are related to language processing in general or some specific part of it.

maybe it will be interesting for them to start from this project and later get better grasp on what is Active Inference is.

awesome totally agree. It's like seeing the range of live streams is really one of the best ways to just jump into understanding the ways and the topics that people speak.

okay.

okay.

we'll just briefly mention the research work happening but.

this last several month period has been extremely focused and extremely productive on education with the two new courses, the qualification system, all of these other curricular developments.

so there's several ongoing research projects.

projects.

we have the active inference ontology as a research and development initiative.

active blockference as a software package for complex systems modeling using active inference.

robotics and embodied with JFs, symbolic active inference robots.

and then there's some project seeds that we always love to offer up to interns or to volunteers as something that they can like.

restart and take a lot of initiative, something that's just planted but barely there.

and that right now is the graphical interface with thinking really expansively about what it looks like to learn and apply active inference.

from an embodied perspective and cognitive agent modeling doing meta analysis and synthesis across different cognitive modeling frameworks.

so that we can really have a place to point when we describe how different cognitive phenomena.

speaking to a cognitive pattern language.

are implemented or addressed by active inference and other frameworks.

and then there's the so many of you who engage in individual research.

so there's a lot of research happening in the space.

and we see education as a fundamental enablement for research.

as part of the Institute's mission for education and research development.

so it's been good times with research.

and I'm really excited to see all the projects that people share during sessions.

start to develop and flourish.

anything else you want to add about research, Alex?

yeah, maybe this direction was not so active last times.

but as we discussed for future directions,

that we will bring more attention and more efforts

to maybe formalize a bit how we as Institute

and service people around from ecosystems,

how we can support them on their ways and so on.

so it will be good times here too.

yeah, agreed.

we want to provide a lot of research services.

so for people who, who want to be on any side of that table, reach out.

all right.

well, the last section here is going to be a short active inference based presidential address.

i will be just speaking to some recent developments as well as past developments in the active inference space.

and just give a few thoughts on how they relate to the active inference Institute and what we're all doing here.

if you're watching live, please write some questions or comments into the live chat.

and at the end of this short address, we will review any other comments or questions from those who are here live.

give some last thoughts and then we'll be back into the work.

so i wanted to begin this address with a reference to the work that was before active inference Institute, which was active inference lab.

but even before active inference lab,

there was the collaboration with Alex, Yvonne, Alexandra, RJ and myself

around active inference and behavior engineering for teams and organizations.

and this was in September, 2020,

only a few years ago now, more than two, but less than three years ago.

and it's amazing how far things have come.

and at the same time, how many of the topics that we were able to bring into this context of remote teams and organizations

still ring as relevance.

and they're just gestured to here.

we've gone far towards enacting and implementing and improving,

but still we can think about cognitive entities engaged in the cyber physical niche with narrative and rhetoric, structuring their understanding of how sensory data translate into action.

So for people who are interested in this topic,

we invite you one way or another to engage with us on this journey,

because we are only going to do it all together.

So just wanted to begin with that kind of callback to several years ago,
though so much has changed certainly for all of us in the times between,
that when we really find first principles approaches like active inference,
and when we really care about systems of interest and communities,
then the actions that we take, we don't regret.

But I think we can go into a little bit more detail and reference some recent developments in the active inference ecosystem

that speak to how day after day, week after week,
our system of interest, active inference, learning and application,
really gifts or blesses us with new ways of thinking and new ways of acting.

So for example, just within the last months,

the inner screen model of consciousness was released as a preprint by Maxwell Ramstead at all,
dealing with all these topics that we know and love and building upon the quantum free energy principle formalism,

which again, it's easy to forget was not even in place two years ago,
discussed in live stream number 40.

And weeks after publication,

we're able to enact a conversation,

hearing from voices all over the world,

bringing experts and those with different backgrounds onto the screen.

Here, the screen that you see is the classical information depository.

That enables us as quantum cognitive agents to engage with each other.

And so all of these beautiful people on the stream,

who with their regime of attention and their quantum cognitive rotation,

brought to bear their attention and their language contribution through action.

We are able to inscribe that on classical digital artifacts like live streams and transcripts,

and then provide it to the niche for all of you to rotate that into your own space and context.

So again, the spatial temporal and conceptual linkages are happening at a speed that is boggling.

And on the right is a paper from just this month,

officially by Thomas Parr at all cognitive effort and active inference.

And I have the image small because there's more details to go into in the paper.

However, they used a well-known model of cognitive burden called the Stroop model,
which shows a color written out as a word.

And that word is shown in some color.

So the color red written in red, the color red written in blue,

and the timing dynamics of people's reading behavior and recognition behavior is a proxy for cognitive effort.

And so this is just one more, not even brick in the wall,

not even just one tool in the toolkit,

but like another lens or layer or ability or capacity for active inference framework,

which adds these features, affect, memory, anticipation.

All these different features are layered in like a symphony without developing bloat,

because the first principles kernel, the cognitive firmware is not getting bloated.

It is staying at first principles.

And sometimes it feels like even diving down to zero with principles.

While advanced research in the space is able to help us understand

really sophisticated cognitive phenomena like cognitive effort.

Now one can look back through the previous stream and what we discussed,

and how could cognitive effort help us support the accessibility, rigor,

and applicability of active inference.

It's such a natural question.

And again, week after week, our niche provides us with new ways of thinking and doing.

So it's so fun to be gripping and gripped within active inference.

I'd like to recognize and commend Jakob's excellent work.

Just in April, we published generalized notation notation for active inference models.

GNN, generalized notation notation is a plain text format for conveying active inference generative models.

It's something like a dialect of markdown.

And it brings us closer and closer to the triple play.

The triple play is the idea that we should have integrity amongst multiple representations of active inference models.

That is, what we see visualized graphically or tangibly,

what we hear and say in terms of the natural language use,

and with executable cognitive simulations.

So if someone looks at an organizational flow chart,

and there's three A's connected to two B's,

that is something that we can visualize, describe, and execute,

so that there isn't a front and a back room conversation.

Rather, there's an integrated conversation that allows for everybody to make contributions in the modalities and in the ways that they want.

And the interlingua, not the only interlingua, and by no means its final version,

is the plain text generalized notation notation.

We're having a lot of fun working with language models and visualizing this,

transforming it into executable simulations.

And so thanks again, Jakob, for the really creative and dedicated work,

because this method is going to be a real core functionality for translating across visual elements,

our natural languages, and the kinds of simulations that we want to run.

And I'll close with a recent publication with Karl Friston et al., including myself and Blue.

So two out of three officers on the paper, not terrible.

And in this paper, we used free energy principle and active inference

to provide something like a measuring or an observability framework for ecology, evolution, and development, or eco-evo-devo.

And the way that this was done is through the renormalization group of physics.

And so in this image on the left, we can see it as like maybe a bunch of cells,

which each contain organelles, and those cells make a tissue.

Or the big circle might be the population of organisms or the ecosystem community.

Again, recognizing the nested niche engagement within and across levels of spatial and temporal modeling.

And looking at just two layers and how they're connected, we can think of that as a nested action perception cycle.

So here's like the behavior, sensory, and the behavior of a worm, in this case, happening at whatever timescale it's being modeled at.

And then at the slower or the deeper level of the hierarchical model, we have the Bayesian learning-like process of the extended genotype happening in a slower adaptive layer.

And so this should clearly connect to what we said at the very beginning of the stream about the institute as part of a nested system.

Here we have the ecosystem, the space around.

Here's the institute.

This is what it really means to enter and to join.

Not just to look into, but to be on or in the institute.

And then the kinds of dynamic partitions that we're able to embody within the institute through our communications and through our actions.

And so it's just so exciting to see everyone's incredible research contributions, the kinds of questions and developments that even if I were not an author, even if I were not in the active inference field, I would believe are at least the right things to be trying in terms of bringing theoretical integration in ecology, evolution, and development.

At least these are the right research avenues.

But then again, to be able to act, infer, and serve through the institute as a multi-scale evolutionary entity in and of itself.

This kind of real-time enactment of multiple scales and quantum reference frames is something that is only arising through co-creation.

So, in closing, I send my deepest appreciation to all those who support and care and engage.

To those who are not sure, I welcome you.

To the past, the present, and the future, we respect.

And that's why we do what we do.

So, Alex, where do we go or what is next?

Yeah, thank you very much for the talk.

And really providing that vision from the beginning and how it's going now.

And congratulations on that publication on evolutionary work.

And it's a great opening direction.

And I believe this direction will make a next step into techno-evolution also.

And we will use this techno-evolution as a principle approach to develop an institute and a ecosystem.

And that's the key to this.

Yes.

I'll just wait a few more seconds if anyone writes a question that we can address in the live chat.

Let's look at some of these questions that we look at at the end of every live stream.

What would a good understanding enable?

Well, if we were able to really understand our own generative model as the institute and as the active inference ecosystem,

it would support our sense-making and decision-making.

We'd be able to act and infer and serve better.

What are the unique predictions and implications?

That's a very interesting question.

I'm sure there are many ways to address this.

And we did have many future-oriented expressions during this discussion even today.

What are the next steps for free energy principle and active inference?

Again, do you have any thoughts on this one, the middle question or the second question, Alex?

I believe that the unique predictions and next steps depends on some specific field or domain.

For ourselves, we understand that active inference is a scale-free approach and it can be applied to any scale of research,

any domain which any researcher or engineer is working on.

It can be described with active inference language and starting from where?

Using the performances to computational modeling or from our types of modeling and so on.

Yeah.

What are the goals of the research?

To embody our mission to support the active inference ecosystem, to be the best institute we can be, be the best nest mates within it.

What do you think?

Maybe we should remove goals.

Let's change it to preferences.

Yeah, at least.

Yeah, at least.

And considering it in open-endedness way, again, we can speak about goals, but it looks like it's not so useful concept.

Yes.

Sometimes when we're talking about the specific research papers, we talk about the goal of that paper, but goal is not to be found in the active inference ontology.

And we find ways of making do and thriving without needing to rest upon goals and reward.

And that's really exciting.

And that's something that a lot of people don't even realize that they're steeped in reward maximization and goal implementation until the open-endedness and the vitality of active inference becomes understood.

What are you still curious about?

I'll let you go first, then I'll give a last thought as we close.

I always remember our discussions about sense of the word curious in the beginning.

I believe that we're more and more curious with each week, each day, each month, what our field, our frontier are showing us so much.

A lot of different perspectives, different connections, real people in communications and projects and so on.

So it's happiness.

Very beautiful.

I'm curious about many things like that you said that we develop and hone that curiosity every day and we seek to spark and be sparked.

When people come in from different backgrounds with their curiosities, it's inspiring to say the least.

And I'm really curious about who will have listened or read this far in the stream and choose that moment to take an action to get involved with the Institute.

So there's only one way that my uncertainty is going to be reduced about that.

And I think if you know, you know.

So thank you, Alex.

Thank you, Blue.

Thank you to the officers and to all of the Institute.

See you around.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.